

The Second Law of Thermodynamics

PHYS 204 · Lecture 5 · Heat engines, entropy and efficiency

1. Why a second law?

The first law tells us that energy is conserved: heat added to a system either raises its internal energy or is turned into work. It says nothing, however, about the direction in which natural processes run. A hot cup of tea cools down in a room; it never spontaneously draws heat from the colder air and becomes hotter. The second law of thermodynamics captures exactly this one-way character of nature.

There are two classical statements of the law. The Kelvin-Planck statement says that no cyclic device can convert heat from a single reservoir entirely into work. The Clausius statement says that heat cannot flow by itself from a colder body to a hotter one. The two statements are equivalent: violating one would let us build a machine that violates the other.

2. Heat engines

A heat engine takes in heat Q_H from a hot reservoir, delivers useful work W , and rejects the remaining heat Q_C to a cold reservoir. Because the engine returns to its initial state after each cycle, its internal energy does not change, and the first law gives the energy balance shown in Figure 1.

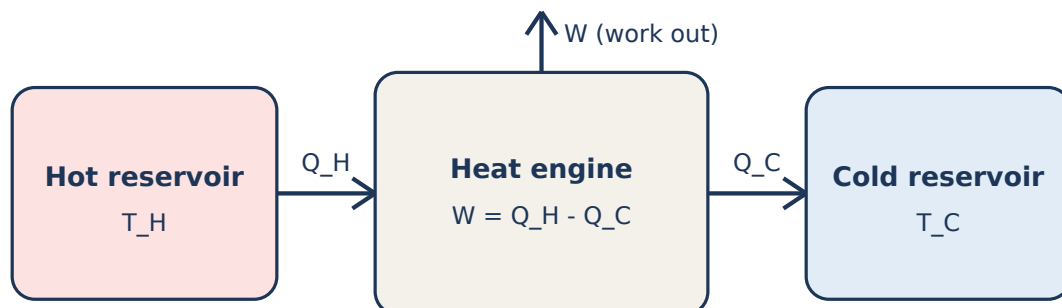


Figure 1. Energy flow in a heat engine.

The thermal efficiency is the fraction of the input heat that is turned into work:

$$\eta = W / Q_H = 1 - Q_C / Q_H$$

Real engines are far from perfect. A car engine typically reaches 25–35 %, a modern combined-cycle power plant around 60 %. The second law tells us that even an ideal engine can never reach 100 %.

3. The Carnot engine

Sadi Carnot showed in 1824 that the most efficient engine operating between two temperatures is a reversible one. Its cycle consists of two isothermal and two adiabatic steps, and its efficiency depends only on the reservoir temperatures (in kelvin):

$$\eta_{\text{Carnot}} = 1 - T_{\text{C}} / T_{\text{H}}$$

Step	Process	Heat	Work
1 → 2	Isothermal expansion at T_{H}	Q_{H} in	by gas
2 → 3	Adiabatic expansion	0	by gas
3 → 4	Isothermal compression at T_{C}	Q_{C} out	on gas
4 → 1	Adiabatic compression	0	on gas

Example. A steam turbine runs between 800 K and 300 K. Its maximum possible efficiency is $\eta = 1 - 300/800 = 0.625$, or 62.5 %. Any real turbine between these temperatures will do worse, because friction and finite-speed heat transfer make its cycle irreversible.

4. Entropy

Clausius introduced a new state function, entropy, to express the second law quantitatively. For a reversible process, the change in entropy is the heat transferred divided by the absolute temperature at which it is transferred:

$$\Delta S = \int dQ_{\text{rev}} / T$$

For an isolated system, the total entropy never decreases: $\Delta S \geq 0$. It stays constant only for idealised reversible processes and grows in every real one. This is the most general form of the second law, and it is why we speak of an arrow of time.

Key ideas to remember

- Heat flows spontaneously only from hot to cold.
- No engine can be more efficient than a Carnot engine between the same temperatures.
- The entropy of an isolated system never decreases.

5. Practice problems

Problem 1. An engine absorbs 2 000 J of heat from a reservoir at 600 K and rejects 1 200 J to a reservoir at 300 K. (a) What is its efficiency? (b) Compare it with the Carnot efficiency for the same reservoirs.

Problem 2. One kilogram of ice melts at 0 °C. The latent heat of fusion is 334 kJ/kg. Find the change in entropy of the ice.

Problem 3. Explain in your own words why a refrigerator does not violate the Clausius statement of the second law.